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Device housing features to facilitate damage free guidewire translations

Abstract

Various aspects of the present disclosure are directed towards apparatuses, systems, and methods that include a percutaneous circulatory support device. The percutaneous circulatory support device is used with a guidewire and may include a housing having a blood outlet aperture. The blood outlet aperture may include a channel that is configured to receive and support the guidewire.

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Background/Summary

CROSS REFERENCE TO RELATED APPLICATION (1) This application claims priority to Provisional Application No. 63/280,345, filed Nov. 17, 2021, which is herein incorporated by reference in its entirety.

TECHNICAL FIELD

(1) The present disclosure relates to percutaneous circulatory support systems. More specifically, the disclosure relates to percutaneous circulatory support devices that are delivered to a patient's heart using a guidewire.

BACKGROUND

(2) Percutaneous circulatory support devices can provide transient support for up to approximately several weeks in patients with compromised heart function or cardiac output. Such devices are typically delivered to a patient's heart using a guidewire, whereby the circulatory support device is coupled to and moved along the guidewire through the patient's vasculature until the device is in the proper position within the heart. Generally, circulatory support devices are coupled to a guidewire by feeding the proximal end of a guidewire, the distal end of which has already been inserted into a patient's vasculature, through an opening in the device. The proximal end of the guidewire is then passed through the device and exits the device via another opening. The device is then able to be moved along the length of guidewire to insert the device into the patient's vasculature and ultimately into the patient's heart. As the guidewire passes through the openings in the device or the device moves along the guidewire, the guidewire may be damaged because the device is typically made of a metal material, while the guidewire is composed of a softer material or contains a coating. For example, the coating of the guidewire may be removed by contact between the guidewire and a surface of one of the openings. The removal of the coating material may cause health complications or negatively affect the performance of the guidewire and/or device.

SUMMARY

(3) In an Example 1, a percutaneous circulatory support device for use with a guidewire comprises a housing including an interior lumen, an exterior surface, a blood inlet, and a blood outlet that includes a blood outlet aperture, wherein the blood outlet aperture includes a channel extending from the interior lumen to the exterior surface of the housing, the channel configured to non-marringly receive and support the guidewire.

(4) In an Example 2, the percutaneous circulatory support device of Example 1, the device further comprises an impeller disposed within the interior lumen, the impeller configured to rotate relative to the housing to cause blood to flow into the blood inlet, through the interior lumen of the housing, and out of the blood outlet, and a motor operatively coupled to the impeller, the motor configured to rotatably drive the impeller.

(5) In an Example 3, the percutaneous circulatory support device of Example 1 or 2, wherein the blood outlet aperture is a first blood outlet aperture, the blood outlet includes a second blood outlet aperture, and the channel extends proximally beyond the second blood outlet aperture.

(6) In an Example 4, the percutaneous circulatory support device of any one of Examples 1 to 3, wherein the channel includes a first end portion which is sloped.

(7) In an Example 5, the percutaneous circulatory support device of any one of Examples 1 to 3, wherein the channel includes a first end portion which is rounded.

(8) In an Example 6, the percutaneous circulatory support device of any one of Examples 1 to 5, wherein the channel includes a width greater than the guidewire.

(9) In an Example 7, the percutaneous circulatory support device of any one of Examples 1 to 6, wherein the channel has a length of 0.025 to 0.375 inches.

(10) In an Example 8, the percutaneous circulatory support device of any one of Examples 1 to 7, wherein the channel includes a substantially flat surface to support the guidewire.

(11) In an Example 9, the percutaneous circulatory support device of any one of Examples 1 to 8, wherein the channel is located on a proximal portion of the blood outlet aperture.

(12) In an Example 10, the percutaneous circulatory support device of any one of Examples 1 to 9, wherein the housing further comprises a first housing portion proximal of the channel, a second

housing portion distal of the channel, wherein the first housing portion has a first diameter and the second housing portion has a second diameter, the first diameter being smaller than the second diameter, and a tapered housing portion that extends between the first housing portion and the second housing portion, the channel being located within the tapered housing portion.

(13) In an Example 11, the percutaneous circulatory support device of Example 10, wherein the channel includes a first side wall and a second side wall, the first and second side walls configured to form a flared opening adjacent the first housing portion.

(14) In an Example 12, a method for using a percutaneous circulatory support device comprises inserting a distal end of a guidewire into the vasculature of a patient, inserting a proximal end of the guidewire into a housing of the device, the housing comprising an interior lumen, an impeller within the interior lumen, an exterior surface, and a blood outlet that includes a blood outlet aperture, the blood outlet aperture includes a channel, the channel extending from the interior lumen to the exterior surface of the housing and configured to non-marringly receive and support the guidewire, passing the proximal end of the guidewire through the interior lumen of the housing, passing the proximal end of the guidewire adjacent to the impeller, and passing the proximal end of the guidewire through the channel of the blood outlet aperture such that the guidewire extends from the interior lumen to an exterior surface of the housing.

(15) In an Example 13, the method of Example 12, wherein the channel is located on a proximal portion of the blood outlet aperture.

(16) In an Example 14, the method of Example 13, wherein the blood outlet aperture is a first blood outlet aperture, the housing includes a second blood outlet aperture, and the channel extends proximally beyond the second blood outlet aperture.

(17) In an Example 15, the method of any one of Examples 12 to 14, further comprising moving the device along the guidewire and inserting the device into the vasculature of the patient.

(18) In an Example 16, a percutaneous circulatory support device for use with a guidewire, comprises a housing comprising an interior lumen, an exterior surface, a blood inlet, and a blood outlet that includes a blood outlet aperture, an impeller disposed within the interior lumen, the impeller configured to rotate relative to the housing to cause blood to flow into the blood inlet, through the interior lumen of the housing, and out of the blood outlet; and a motor operatively coupled to the impeller, the motor configured to rotatably drive the impeller, wherein the blood outlet aperture includes a channel extending from the interior lumen to the exterior surface of the housing, the channel configured to non-marringly receive and support the guidewire.

(19) In an Example 17, the percutaneous circulatory support device of Example 16, wherein the blood outlet aperture is a first blood outlet aperture, the blood outlet includes a second blood outlet aperture, and the channel extends proximally beyond the second blood outlet aperture.

(20) In an Example 18, the percutaneous circulatory support device of Example 16, wherein the channel includes a first end portion which is sloped.

(21) In an Example 19, the percutaneous circulatory support device of Example 16, wherein the channel includes a first end portion which is rounded.

(22) In an Example 20, the percutaneous circulatory support device of Example 16, wherein the channel includes a width greater than the guidewire.

(23) In an Example 21, the percutaneous circulatory support device of Example 16, wherein the channel has a length of 0.025 to 0.375 inches.

(24) In an Example 22, the percutaneous circulatory support device of Example 16, wherein the channel includes a substantially flat surface to support the guidewire.

(25) In an Example 23, the percutaneous circulatory support device of Example 16, wherein the channel is located on a proximal portion of the blood outlet aperture.

(26) In an Example 24, the percutaneous circulatory support device of Example 16, wherein the housing further comprises a first housing portion proximal of the channel, a second housing portion distal of the channel, wherein the first housing portion has a first diameter and the second housing

portion has a second diameter, the first diameter being smaller than the second diameter, and a tapered housing portion that extends between the first housing portion and the second housing portion, the channel being located within the tapered housing portion.

(27) In an Example 25, the percutaneous circulatory support device of Example 24, wherein the channel includes a first side wall and a second side wall, the first and second side walls configured to form a flared opening adjacent the first housing portion.

(28) In an Example 26, a percutaneous circulatory support device for use with a guidewire comprises a housing comprising an interior lumen, an exterior surface, a blood inlet, and a blood outlet that includes a first blood outlet aperture and a second blood outlet aperture, an impeller disposed within the interior lumen, the impeller configured to rotate relative to the housing to cause blood to flow into the blood inlet, through the interior lumen of the housing, and out of the first and second blood outlet apertures, and a motor operatively coupled to the impeller, the motor configured to rotatably drive the impeller, wherein the first blood outlet aperture that includes a channel configured to non-marringly receive and support the guidewire, the channel extending from the interior lumen to the exterior surface of the housing and proximally beyond the second blood outlet aperture.

(29) In an Example 27, the percutaneous circulatory support system of Example 26, wherein the channel includes a first end portion which is sloped.

(30) In an Example 28, the percutaneous circulatory support system of Example 26, wherein the channel includes a first end portion which is rounded.

(31) In an Example 29, the percutaneous circulatory support system of Example 26, wherein the channel includes a substantially flat surface to support the guidewire.

(32) In an Example 30, the percutaneous circulatory support system of Example 26, wherein the housing further comprises a first housing portion proximal of the channel, a second housing portion distal of the channel, wherein the first housing portion has a first diameter and the second housing portion has a second diameter, the first diameter being smaller than the second diameter, and a tapered housing portion that extends between the first housing portion and the second housing portion, the channel being located within the tapered housing portion.

(33) In an Example 31, the percutaneous circulatory support device of Example 30, wherein the channel includes a first side wall and a second side wall, the first and second side walls configured to form a flared opening adjacent the first housing portion.

(34) In an Example 32, a method for using a percutaneous circulatory support device, comprises inserting a distal end of a guidewire into the vasculature of a patient, inserting a proximal end of the guidewire into a housing of the device, the housing comprising an interior lumen, an impeller within the interior lumen, an exterior surface, and a blood outlet that includes a blood outlet aperture, the blood outlet aperture includes a channel, the channel extending from the interior lumen to the exterior surface of the housing and configured to non-marringly receive and support the guidewire, passing the proximal end of the guidewire through the interior lumen of the housing, passing the proximal end of the guidewire adjacent to the impeller, and passing the proximal end of the guidewire through the channel of the blood outlet aperture such that the guidewire extends from the interior lumen to an exterior surface of the housing.

(35) In an Example 33, the method of Example 32, wherein the channel is located on a proximal portion of the blood outlet aperture.

(36) In an Example 34, the method of Example 33, wherein the blood outlet aperture is a first blood outlet aperture, the housing includes a second blood outlet aperture, and the channel extends proximally beyond the second blood outlet aperture.

(37) In an Example 35, the method of Example 32, further comprising moving the device along the guidewire and inserting the device into the vasculature of the patient.

(38) While multiple embodiments are disclosed, still other embodiments of the present invention will become apparent to those skilled in the art from the following detailed description, which

shows and describes illustrative embodiments of the invention. Accordingly, the drawings and detailed description are to be regarded as illustrative in nature and not restrictive.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

- (1) FIG. 1 is a side sectional view of an illustrative mechanical circulatory support device (also referred to herein, interchangeably, as a “blood pump”), in accordance with embodiments of the subject matter disclosed herein.
- (2) FIG. 2A is a perspective view of the mechanical circulatory support device of FIG. 1, in accordance with embodiments of the subject matter disclosed herein.
- (3) FIG. 2B is a perspective view of an illustrative blood outlet aperture and channel, in accordance with embodiments of the subject matter disclosed herein.
- (4) FIG. 2C is a perspective view of an illustrative blood outlet aperture and channel, in accordance with embodiments of the subject matter disclosed herein.
- (5) FIG. 3 is a perspective view of the mechanical circulatory support device of FIG. 1 and further illustrating a guidewire, in accordance with embodiments of the subject matter disclosed herein
- (6) FIG. 4 is a side sectional view of the mechanical circulatory support device of FIG. 1 further illustrating a guidewire, introducer sheath, and blood vessel.
- (7) While the invention is amenable to various modifications and alternative forms, specific embodiments have been shown by way of example in the drawings and are described in detail below. The intention, however, is not to limit the invention to the particular embodiments described. On the contrary, the invention is intended to cover all modifications, equivalents, and alternatives falling within the scope of the invention as defined by the appended claims.

DETAILED DESCRIPTION

- (8) FIG. 1 depicts a partial side sectional view of an illustrative percutaneous circulatory support device **100** (also referred to herein, interchangeably, as a “blood pump”) in accordance with embodiments of the subject matter disclosed herein. The device **100** may form part of a percutaneous circulatory support system, together with a guidewire (shown elsewhere) and an introducer sheath (shown elsewhere). More specifically, the guidewire and the introducer sheath may facilitate percutaneously delivering the device **100** to a target location within a patient, such as within the patient's heart. Alternatively, the device **100** may be delivered to a different target location within a patient.
- (9) With continued reference to FIG. 1, the device **100** generally includes a housing **101** that includes an impeller housing **102** and a motor housing **104**. In some embodiments, the impeller housing **102** and the motor housing **104** may be integrally or monolithically constructed. In other embodiments, the impeller housing **102** and the motor housing **104** may be separate components configured to be removably or permanently coupled. In some embodiments, the blood pump **100** may lack a separate motor housing **104** and the impeller housing **102** may be coupled directly to the motor **105** described below, or the motor housing **104** may be integrally constructed with the motor **105** described below. Impeller housing **102** includes an interior lumen **102a**, an interior surface **102b**, and an exterior surface **102c**. Similarly, the motor housing **104** includes an interior lumen **104a**, an interior surface **104b**, and an exterior surface **104c**. Center axis **103** passes longitudinally through the housing **101**.
- (10) The impeller housing **102** carries an impeller assembly **106** therein. The impeller assembly **106** includes an impeller shaft **108** that is rotatably supported by at least one bearing, such as a bearing **110**. The impeller assembly **106** also includes an impeller **112** that rotates relative to the impeller housing **102** to drive blood through the device **100**. More specifically, the impeller **112** causes blood to flow from a blood inlet **114** formed on the impeller housing **102**, through the

impeller housing **102**, and out of a blood outlet **116** formed on the impeller housing **102**. In some embodiments and as illustrated, the impeller shaft **108** and the impeller **112** may be separate components, and in other embodiments the impeller shaft **108** and the impeller **112** may be integrated. In some embodiments and as described in more detail below, the inlet **114** and/or the outlet **116** may each include multiple apertures (see e.g., FIGS. 2-4). In other embodiments, the inlet **114** and/or the outlet **116** may each include a single aperture. In some embodiments and as illustrated, the inlet **114** may be formed on an end portion of the impeller housing **102** and the outlet **116** may be formed on a side portion of the impeller housing **102**. In other embodiments, the inlet **114** and/or the outlet **116** may be formed on other portions of the impeller housing **102**. In some embodiments, the impeller housing **102** may couple to a distally extending cannula (not shown), and the cannula may receive and deliver blood to the inlet **114**.

(11) With continued reference to FIG. 1, the motor housing **104** carries a motor **105**, and the motor **105** is configured to rotatably drive the impeller **112** relative to the impeller housing **102**. In the illustrated embodiment, the motor **105** rotates a drive shaft **120**, which is coupled to a driving magnet **122**. Rotation of the driving magnet **122** causes rotation of a driven magnet **124**, which is connected to and rotates together with the impeller assembly **106**. More specifically, in embodiments incorporating the impeller shaft **108**, the impeller shaft **108** and the impeller **112** are configured to rotate with the driven magnet **124**. In other embodiments, the motor **105** may couple to the impeller assembly **106** via other components.

(12) In some embodiments, a controller (not shown) may be operably coupled to the motor **105** and configured to control the motor **105**. In some embodiments, the controller may be disposed within the motor housing **104**. In other embodiments, the controller may be disposed outside of the motor housing **104** (for example, in an independent housing, etc.). In some embodiments, the controller may include multiple components, one or more of which may be disposed within the motor housing **104**. According to embodiments, the controller may be, may include, or may be included in one or more Field Programmable Gate Arrays (FPGAs), one or more Programmable Logic Devices (PLDs), one or more Complex PLDs (CPLDs), one or more custom Application Specific Integrated Circuits (ASICs), one or more dedicated processors (e.g., microprocessors), one or more Central Processing Units (CPUs), software, hardware, firmware, or any combination of these and/or other components. Although the controller is referred to herein in the singular, the controller may be implemented in multiple instances, distributed across multiple computing devices, instantiated within multiple virtual machines, and/or the like. In other embodiments, the motor **105** may be controlled in other manners.

(13) With continued reference to FIG. 1 and additional reference to FIG. 2A, the device **100** facilitates passage of a guidewire (shown elsewhere) through device **100** by the incorporation of a channel **126** in a blood outlet aperture **128** of the blood outlet **116**. As described above, the blood outlet **116** may include more than one aperture, for example as shown in FIG. 2A, the blood outlet **116** includes six blood outlet apertures **128**, **130**, **131**, **132**, **133**, and **134**. As shown in FIG. 2A, the blood outlet aperture **128** includes a proximal portion **136** and a distal portion **138**. In some embodiments and as shown in FIG. 2A, the channel **126** is located on the proximal portion **136** of the blood outlet aperture **128**. In other embodiments, the channel **126** could be located on other portions of a blood outlet aperture, for example on the distal portion **138**. As shown in FIG. 2A, the channel **126** extends from the blood outlet aperture **128** in a proximal direction. As also shown in FIG. 2A, the channel **126** extends proximally beyond each of the other blood outlet apertures **130**, **131**, **132**, **133**, and **134**, thus indicating to a user that a guidewire should be passed through the blood outlet aperture **128** rather than the other blood outlet apertures **130**, **131**, **132**, **133**, and **134**. Additional indicators, such as markings or lettering, may be placed near the channel **126** to indicate the proper location of the guidewire. The channel **126** has a width **140** and a length **142**. The width **140** is generally no narrower than the width of the guidewire. In some embodiments, the length **142** may vary from 0.025 inches to 0.375 inches.

(14) In the embodiment in FIGS. 2A-C, the channel **126** includes a first end portion **144** at the distal end **146** of the channel **126**, that is, where the channel **126** meets interior surface **102b**. Similarly, in the embodiment in FIGS. 2A-C, the channel **126** also includes a second end portion **148** at the proximal end **150** of the channel **126**, where the channel **126** meets the exterior surface **104c**. FIG. 2B illustrates a first end portion **144b** that has not been sloped, beveled, rounded, or otherwise mechanically altered to smooth the first end portion **144**. In contrast, in some embodiments, the first end portion **144** and/or second end portion **148** may be sloped, beveled, rounded, or otherwise mechanically altered to smooth the first end portion **144** and second end portion **148**. For example, as shown in FIG. 2C, first end portion **144c** is rounded to smooth the first end portion **144c**. In some embodiments, channel **126** may include fewer or additional edges or surfaces than shown in FIGS. 2A-C, and some or all of such edges or surfaces may be smoothed through sloping, beveling, rounding, or other mechanical alteration. As further shown in FIGS. 2A-C, in some embodiments, the channel **126** includes a first side wall **152** and a second side wall **154**, separated by the width **140**, both of which extend along the length **142** and flare away from each other to form a flared opening **158** adjacent to the distal end **146** of the channel **126**. The flared opening **158** may be, for example, at most 120 degrees wide. In some embodiments, the first side wall **152** and the second wall **154** may be sloped, beveled, rounded, or otherwise mechanically altered to smooth the surfaces of the first side wall **152** and a second side wall **154**. For example, as shown in FIG. 2C, the second wall **154c** is rounded. In other embodiments, the first side wall **152** and the second wall **154** may remain unaltered, as shown by second wall **154b** in FIG. 2B.

(15) As shown in FIGS. 2A-C and 3, the channel **126** also includes a substantially flat surface **156** extending between the end portions **144**, **148** and the side walls **152**, **154**. Although the surface **156** may be slightly angled or tapered from the proximal end **150** and the distal end **146**, the surface **156** is substantially flat to support the guidewire **200** along the channel **126** in the embodiment shown in FIG. 2 and FIG. 3. The guidewire **200** includes a distal end (not shown) and a proximal end **202**, and may be composed of one or more metals, one or more plastics, composites, or the like. In some embodiments, the guidewire **200** may include a coating (not shown) that facilitates trackability within patient vasculature and reduces friction while advancing the device **100** over the guidewire **200** or while advancing the guidewire **200** through device **100**. Such coatings may include PTFE, polymer coatings, or ceramic coatings through chemical vapor deposition processes such as atomic layer deposition.

(16) In traditional circulatory support devices, as a guidewire is passed through the openings a circulatory support device, or the device moves along a guidewire, the guidewire may be damaged by contact between the guidewire and the device housing. For example, if the housing is made of a metal material, and the guidewire is composed of a softer material or contains a coating, the guidewire may be damaged, or the coating on the guidewire may be scraped off, during contact with relatively abrupt housing surfaces. Also, in traditional circulatory support devices, the guidewire may be bent at a relatively large angle or angles as the guidewire transitions from the interior to the exterior of the device housing.

(17) In contrast to traditional devices, the likelihood of damage is substantially decreased with the device **100** due to the guidewire **200** passing through the channel **126**. In particular, the above-described features of the channel **126** substantially reduce the chances that a coating on the guidewire **200** will be scraped off or damaged. In addition, the above-described features of the channel **126** minimize the amount that the guidewire **200** is deformed or bent as it transitions from the interior lumen **102a** of the impeller housing **102** to the exterior surface **104c** of the motor housing **104**, providing the guidewire **200** with a flatter profile, and thus advantageously taking up less space within the introducer sheath **300**, as shown in FIG. 4. Also, the flatter profile of the guidewire **200** facilitates easier movement of the device **100** along the guidewire **200**, including through the introducer sheath **300** and the blood vessel V.

(18) The above-described features of the channel **126** facilitate non-marringly receiving and

supporting the guidewire **200** as the guidewire **200** passes through the blood outlet aperture **128**. In particular, the channel **126** in the blood outlet aperture **128** provides a relatively smooth surface for the guidewire **200** to pass from the interior lumen **102a** of the impeller housing **102** to the exterior surface **104c** of the motor housing **104**, as shown in FIG. **3**. By incorporating the sloped, beveled, rounded, or otherwise modified first and second end portions **144**, **148**, the first and second side walls **152**, **154**, as well as the flared opening **152** and the substantially flat surface **156**, the likelihood that the guidewire **200** will be damaged as the guidewire **200** passes through the blood outlet aperture **128**, or when the device **100** is moved along the guidewire **200**, is diminished as compared to other blood outlet apertures, such as the blood outlet apertures **130**, **131**, **132**, **133**, and **134**, that do not include such features. For example, the channel **126** reduces the likelihood that a coating on the surface of the guidewire **200** is scraped off when the guidewire **200** passes through the blood outlet aperture **128**, or when the device **100** is moved along the guidewire **200**.

(19) In addition, the above-described features of the channel **126** minimize the amount that the guidewire **200** deforms or bends as the guidewire **200** passes from the interior lumen **102a** of the impeller housing **102** to the exterior surface **104c** of the motor housing **104**, as shown in FIG. **3**. Stated another way, by lengthening the transition area from the interior lumen **102a** of the impeller housing **102** to the exterior surface **104c** of the motor housing **104**, the guidewire **200** assumes a flatter profile as compared to a guidewire transitioning in another blood outlet aperture, such as the blood outlet apertures **130**, **131**, **132**, **133**, and **134** that do not include a feature like the channel **126**. As noted above, a flatter profile of the guidewire allows the guidewire to take up less space within the introducer sheath **300** and also permits easier movement of the device **100** along the guidewire **200**.

(20) With continued reference to FIGS. **2A-C** and **3**, in some embodiments, the impeller housing **102** has an outer diameter that is larger than the outer diameter of the motor housing **104**. In alternative embodiments, the outer diameter of the impeller housing **102** may be substantially equal to the outer diameter of the motor housing **104**, for example when impeller housing **102** and motor housing **104** are integrally or monolithically constructed. In still other embodiments, the outer diameter of the motor housing **104** may be larger than the outer diameter of the impeller housing **102**. As shown in FIGS. **2A-C** and **3**, a tapered housing portion **160** extends between the impeller housing **102** and the motor housing **104**. The tapered housing portion **160** creates a smooth transition along the housing **101** from the larger outer diameter of the impeller housing **102** to the relatively smaller outer diameter of the motor housing **104**. As shown in FIG. **2A-C** and FIG. **3**, the channel **126** is located in the tapered housing portion **160** and is configured to facilitate the passage of the guidewire **200** from the interior lumen **102a** of the impeller housing **102** to the exterior surface **104c** of the motor housing **104**. More specifically, in the embodiment shown in FIG. **2A-C** and FIG. **3** the channel **126** is designed to minimize the amount that the guidewire **200** deforms or bends as the guidewire **200** passes from the interior lumen **102a** of the impeller housing **102** to the exterior surface **104c** of the motor housing **104**. Stated differently, and illustrated in FIGS. **3** and **4**, when the channel **126** is incorporated into a device **100** with a tapered housing portion **160**, the guidewire **200** extends along the exterior surface **104c** of the motor housing **104**, passes through the blood outlet aperture **128** and into the interior lumen **102a** of the impeller housing **102**, while maintaining the same or similar distance from the center axis **103** of the housing **101**.

(21) In other embodiments, other portions besides the proximal portion **136** of the blood outlet **128** aperture may include a channel or similar structure for non-marringly receiving and supporting a guidewire. For example, the distal portion **138** of the blood outlet aperture **128** may include a channel or similar structure for non-marringly receiving and supporting a guidewire. In such an embodiment, the interior surface **102b** of the impeller housing **102** at the distal portion **138** may be configured receive a guidewire, such as by the inclusion of a feature similar to the channel **126** described above. In other embodiments, the interior surface **102b** of the impeller housing **102** at the distal portion **138** may be modified, sculpted, rounded, or otherwise configured to non-marringly

receive and support a guidewire. In addition, in other embodiments, portions of the entirety of a blood outlet aperture may be treated with a coating or surface treatment to prevent damage to a guidewire. In still other embodiments, other apertures, such one or more of the apertures of the blood inlet **114**, may include a channel, such as the channel **126**, or similar structure for non-marringly receiving and supporting a guidewire.

(22) FIGS. **2A-C**, **3**, and **4** facilitate explanation of the method by which the blood pump **100** is used with the guidewire **200** and the advantages provided by the above-described features of the housing **101**, including the channel **126**. In an exemplary procedure, an introducer sheath **300** is inserted into the blood vessel (not shown), and the guidewire **200** is inserted through the introducer sheath **300** such that the proximal end **202** of the guidewire **200** is located outside the introducer sheath **300** and outside of the patient's body. The distal end (not shown) of the guidewire **200** is disposed distally from the introducer sheath **300** and located within the patient's vasculature, for example in the left ventricle of the heart. The blood pump **100** is then coupled to the guidewire **200** by feeding the proximal end **202** of the guidewire **200** through an opening in blood pump **100**, such as the blood inlet **114**. The proximal end **202** of the guidewire **200** is then passed through the housing **101**, for example through the interior lumen **102a** of the impeller housing **102**, past the impeller **112**, and exits the blood pump **100** via another opening, such as the blood outlet **116**, and in particular and as shown in FIGS. **3** and **4**, through the blood outlet aperture **128**. As the guidewire **200** passes through the blood outlet aperture **128**, the guidewire **200** is received and supported by the channel **126**, thereby inhibiting damage to the guidewire **200** as it contacts the housing **101**. FIG. **4** in particular illustrates the guidewire **200** extending proximally from the blood outlet aperture **128**, including along the exterior surface **104c** of the motor housing **104** and within the introducer sheath **300** and blood vessel **V**. Once coupled to the guidewire **200**, the blood pump **100** is able to be moved along the length of the guidewire **200**, through the introducer sheath **300** to insert the device **100** into the blood vessel **V** and ultimately into the patient's heart.

(23) Various modifications and additions can be made to the exemplary embodiments discussed without departing from the scope of the present invention. For example, while the embodiments described above refer to particular features, the scope of this invention also includes embodiments having different combinations of features and embodiments that do not include all of the described features. Accordingly, the scope of the present invention is intended to embrace all such alternatives, modifications, and variations as fall within the scope of the claims, together with all equivalents thereof.

Claims

1. A percutaneous circulatory support device for use with a guidewire, comprising: a housing comprising an interior lumen, an exterior surface, a blood inlet, and a blood outlet that includes a blood outlet aperture; an impeller disposed within the interior lumen, the impeller configured to rotate relative to the housing to cause blood to flow into the blood inlet, through the interior lumen of the housing, and out of the blood outlet; and a motor operatively coupled to the impeller, the motor configured to rotatably drive the impeller, wherein the blood outlet aperture includes a channel extending from the interior lumen to the exterior surface of the housing, the channel having a tapered surface along the exterior surface of the housing proximal of the blood outlet aperture, the channel configured to non-marringly receive and support the guidewire.
2. The percutaneous circulatory support device of claim 1, wherein the blood outlet aperture is a first blood outlet aperture, the blood outlet includes a second blood outlet aperture, and the channel extends proximally beyond the second blood outlet aperture.
3. The percutaneous circulatory support device of claim 1, wherein the channel includes a first end portion which is sloped.
4. The percutaneous circulatory support device of claim 1, wherein the channel includes a first end

portion which is rounded.

5. The percutaneous circulatory support device of claim 1, wherein the channel includes a width greater than the guidewire.

6. The percutaneous circulatory support device of claim 1, wherein the channel has a length of 0.025 to 0.375 inches.

7. The percutaneous circulatory support device of claim 1, wherein the channel includes a substantially flat surface to support the guidewire.

8. The percutaneous circulatory support device of claim 1, wherein the channel is located on a proximal portion of the blood outlet aperture.

9. The percutaneous circulatory support device of claim 1, wherein the housing further comprises: a first housing portion proximal of the channel; a second housing portion distal of the channel; wherein the first housing portion has a first diameter and the second housing portion has a second diameter, the first diameter being smaller than the second diameter; and a tapered housing portion that extends between the first housing portion and the second housing portion, the channel being located within the tapered housing portion.

10. The percutaneous circulatory support device of claim 9, wherein the channel includes a first side wall and a second side wall, the first and second side walls configured to form a flared opening adjacent the first housing portion.

11. A percutaneous circulatory support device for use with a guidewire, comprising: a housing comprising an interior lumen, an exterior surface, a blood inlet, and a blood outlet that includes a first blood outlet aperture and a second blood outlet aperture; an impeller disposed within the interior lumen, the impeller configured to rotate relative to the housing to cause blood to flow into the blood inlet, through the interior lumen of the housing, and out of the first and second blood outlet apertures; and a motor operatively coupled to the impeller, the motor configured to rotatably drive the impeller, wherein the first blood outlet aperture includes a channel configured to non-marringly receive and support the guidewire, the channel extending from the interior lumen to the exterior surface of the housing and proximally beyond the second blood outlet aperture, the channel having a tapered surface along the exterior of the housing proximal of the blood outlet aperture.

12. The percutaneous circulatory support system of claim 11, wherein the channel includes a first end portion which is sloped.

13. The percutaneous circulatory support system of claim 11, wherein the channel includes a first end portion which is rounded.

14. The percutaneous circulatory support system of claim 11, wherein the channel includes a substantially flat surface to support the guidewire.

15. The percutaneous circulatory support system of claim 11, wherein the housing further comprises: a first housing portion proximal of the channel; a second housing portion distal of the channel; wherein the first housing portion has a first diameter and the second housing portion has a second diameter, the first diameter being smaller than the second diameter; and a tapered housing portion that extends between the first housing portion and the second housing portion, the channel being located within the tapered housing portion.

16. The percutaneous circulatory support device of claim 15, wherein the channel includes a first side wall and a second side wall, the first and second side walls configured to form a flared opening adjacent the first housing portion.

17. A method for using a percutaneous circulatory support device, comprising: inserting a distal end of a guidewire into a vasculature of a patient; inserting a proximal end of the guidewire into a housing of the device, the housing comprising an interior lumen, an impeller within the interior lumen, an exterior surface, and a blood outlet that includes a blood outlet aperture, the blood outlet aperture includes a channel, the channel extending from the interior lumen to the exterior surface of the housing proximal of the blood outlet aperture, having a tapered surface along the exterior surface of the housing, and configured to non-marringly receive and support the guidewire; passing

the proximal end of the guidewire through the interior lumen of the housing; passing the proximal end of the guidewire adjacent to the impeller; and passing the proximal end of the guidewire through the channel of the blood outlet aperture such that the guidewire extends from the interior lumen to an exterior surface of the housing.

18. The method of claim 17, wherein the channel is located on a proximal portion of the blood outlet aperture.

19. The method of claim 18, wherein the blood outlet aperture is a first blood outlet aperture, the housing includes a second blood outlet aperture, and the channel extends proximally beyond the second blood outlet aperture.

20. The method of claim 17, further comprising moving the device along the guidewire and inserting the device into the vasculature of the patient.
